

A Comparative Analysis of Inkjet-Printed and Fine Metal Mask Patterned OLED Notebook Panels: Benefits, Challenges, and Costs

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Abstract

This paper evaluates the benefits and challenges of inkjet-printed (IJP) versus conventional fine metal mask (FMM) patterned OLED panels. It analyzes the manufacturing total costs of 16.3-inch notebook panels by comparing materials, equipment depreciation, yield rates, and other parameters to evaluate the trade-offs between the two fabrication approaches.

Author Keywords

RGB; OLED; Fine Metal Mask (FMM); Inkjet Printing (IJP); notebook; hybrid; Gen 8.6; Vacuum Thermal Evaporation (VTE); yield

1. Introduction

Panel manufacturers and their suppliers have been developing industrial IJP processes, equipment, and materials for a variety of applications for much of the flat panel display (FPD) industry's history. After all, in theory at least, it is an eloquent approach. IJP is very efficient; it directly prints precise patterns of only the required amount of material, it operates in an ambient environment, without vacuum chambers and often without an inert gas chamber, and it does not require photomasks, expensive photolithography tools, related equipment, or processes. These characteristics create the opportunity for significant cost reduction compared to conventional manufacturing approaches.

In practice, though, IJP has proven very difficult to implement in mass production applications. Printing on large, non-absorbent, glass substrates, at high speeds, with high drop location and size accuracy, and high uniformity requires a complicated mix of hardware and software expertise. It is, by many magnitudes, more difficult than printing ink on paper. Additionally, formulating required functional materials into low-viscosity "jettable" inks often negatively affects the material's performance.

There are some success stories, notably, printing of polyimide alignment films in LCDs and organic passivation layers for thin film encapsulation (TFE) on flexible OLEDs. However, it is in the area of fine patterning applications that IJP offers the greatest potential value add, and which is much more challenging.

LCD color filter (CF) patterning was the focus of IJP development for many years. Eventually, a couple of makers did commercialize the application, but IJP failed to become the standard approach because the trade-offs required to commercialize IJP CF led to higher cost and lower quality than originally expected.

In many ways, the inkjet printing of OLEDs, to date, has been a similar story to that of LCD CFs. However, continuous technological improvements and considerably greater benefits in the case of RGB OLEDs have kept the promise alive. Now, with one of the leading FPD manufacturers building the world's first Gen 8.6 mass production factory, IJP is closer to delivering on its promises than ever. If successful, it could potentially disrupt the status quo of FMM-based manufacturing, reduce OLED panel costs, and expand the market for OLED-based FPDs.

2. Methodology

In this study, the total manufacturing cost for a 16.3", 3456 x 2234, hybrid IJP OLED, oxide TFT OLED panel for notebook applications was analyzed and compared to a similar, conventional OLED fabricated using FMM patterning of subpixels.

Table 1. Key cost modeling parameters

Parameter	IJP	FMM
Substrate size	2290 x 2620 mm	2290 x 1310 mm
Time range	2025-2028	2025-2028
Fab operation	90%	90%
Capacity (TFT equivalents)	30,000 substrates per month	30,000 substrates per month
Depreciation	5 years	5 years
Capital costs	Oxide TFT only process + IJP related equipment frontplane + module	Oxide TFT only process + FMM related equipment frontplane + module
Material costs	OLED inks, IJP heads, other	Evaporated OLED materials, FMM, other
Yield curve	74%-84 over range	74%-84 over range

The difference in the capital cost requirements for the two technologies is the key parameter in this comparison.

At the same time, evaluating the cost of consumables, including lifetime and material use efficiency, the frontplane process flow, and panel cutting efficiency, is also required to accurately model manufacturing total costs.

Similar yield rates, production time ranges, fab operation, capacity, TFT backplane process flows and equipment, depreciation periods, etc., are applied to both panel types to provide a "likes" comparison. In reality, these parameters may vary between the technologies, but by applying similar values, the OLED patterning technology-specific cost variations are manifested.

A wide variety of other assumptions and parameters are used to model costs. These are not discussed in detail here as they do not substantially influence the variations in the manufacturing of IJP and FMM-type OLED panels.

Price, cost, yield, and other data are regularly collected through systematic, in-person interview-type surveys of flat panel display (FPD) supply chain companies by a team of analysts based in the regions where FPDs are primarily manufactured. Survey participants include material and component makers, equipment makers, panel makers, set makers/brands, and other relevant companies.

Information gathered is data-based and checked every quarter to validate and track changes over time. These databases enable bottom-up modeling of panel costs.

This standard methodology was used to model conventional FMM OLED panel manufacturing costs.

To model IJP panel costs, targeted surveys were conducted to gather pertinent data and confirm parameters related to IJP OLED manufacturing and costs. This IJP-specific information was used to adjust and expand the standard model to cover IJP OLED panels.

After the model was complete, a second round of surveys was conducted to validate and adjust the results before finalizing the conclusions reported here.

3. Results

IJP RGB OLEDs offer many potential benefits (Table 2). However, up until now, the IJP challenges (Table 3) have limited mass production to low volumes.

Table 2. Advantages of IJP compared to FMM OLED

Detail	Comments
Flexible “digital” production	Reduced development time, no need for FMM, FMM cleaning, FMM tension, and so on. Can change print parameters. Fast development & changeover. Small lot production.
Relatively low equipment costs	Compared with an FMM line. Significantly lower patterning equipment costs
High material utilization	500% more efficient than FMM
Relatively lower maintenance costs	Compared with an FMM line
Relatively higher uptime	Compared with an FMM line
2x or greater aperture ratio	Compared with an FMM patterns
Smaller factory footprint	Compared with an FMM line
Process and equipment improvement	Continuous incremental improvement and learning from previous mistakes enable high yield.
Improved panelization	Full-size Gen 8.6 vs. Half cut FMM, but varies by panel size. Also, multi-model glass (MMG) compatibility
Material improvement and new development	If Chinese material makers are aggressive in pursuing, they will expand small molecule development.
Long-term hope for QDEL material improvements	Transition IJP processes and know-how to QDEL production in the long run

Printing RGB subpixels substantially simplifies the color patterning compared with conventional FMM processes. This is due to fewer vacuum steps, a smaller clean room footprint, no FMM handling and cleaning systems, rapid product development and production changeovers, and so on.

Also, OLED ink utilization is well above 90% as inks are dropped directly into line trenches or pixel wells. Only a small amount of the expensive organic materials is wasted.

On the other hand, in the FMM process, metal masks screen where organic materials are deposited onto the substrate. This means that at least two-thirds of the material lands on the mask. Organic material vapor evaporating up from the source also deposits on

various other parts of the chamber. So, FMM material utilization is much lower than IJP, typically less than 20%.

For similar reasons, direct printing enables larger subpixel aperture ratios than shadow effect-constrained FMM. This facilitates side-by-side RGB subpixels, which are preferable for displaying text compared to sub-pixel rendering techniques commonly applied in FMM OLEDs. It also mitigates some of the IJP OLED ink’s relatively lower performance specifications.

Half-cut frontplane processing is an inherent disadvantage of FMM OLED manufacturing. In the TFT process, substrates are handled at native size. However, owing to FMM sagging and alignment issues, glass is cut in half for the evaporation process. Glass cutting generates particles, reduces process efficiency, induces glass strain, and wastes valuable glass area.

In IJP, even with the few open mask evaporation steps, sagging is irrelevant; substrate alignment can be effectively managed by the printer. Full-size substrate processing is possible, which increases efficiency. The extent of the advantage depends on the substrate size, panel size, aspect ratio, panel orientation, and whether MMG is adopted.

In the 16.3-inch example analyzed in this study, 66 panels up were assumed for IJP, while only 60-up were used for FMM. By not having to cut the mother glass substrate in half for the frontplane process, an economical panel layout is achieved. IJP’s 10% efficiency improvement translates directly to lower manufacturing total costs.

Table 3. Disadvantages of IJP compared to FMM OLED

Detail	Comments
Ink compatibility	Inks must be compatible with print heads, which defines viscosity and what type of solvents can be used. Solvents can negatively affect OLED material performance. Must be thoroughly purified.
Ink incompatibility and the remaining requirement of open mask evaporation	ETL, electron injection layer (EIL), cathode, capping layer (CPL), and charge generation layer (CGL) are currently incompatible with solution printing. Although being developed, commercial viability is unclear.
Material lifetime and efficiency	Particularly blue. Much lower than evaporated blue. Large pixel aperture compensates for this, but cannot compete with tandem FMM.
Tandem is not currently feasible	Lack of printable and robust CGL material
High resolution	Limited by drop size, etc. Recent technology advancements may enable 350 PPI, but with trade-offs.
Various print/dry mura	Equipment and process technology are evolving to mitigate.
Technology immaturity	Unproven in high-volume, competitive market segments

Not all frontplane process materials and steps are compatible with inkjet yet. Despite ongoing development, current ETL, cathode, CGL, and CPL ink materials do not offer sufficient performance. Thus, even in the IJP process, these critical materials are deposited

by open mask VTE.

This evaporator is smaller, simple in configuration, has high productivity, and is low in price compared with FMM VTE tools. Even so, it does complicate the IJP process and represents a substantial capital cost.

Material efficiency and thickness requirements may cause higher power consumption, particularly at higher resolutions.

Similarly, no IJP-appropriate CGL materials are yet commercially available. Evaporated CGL films will degrade rapidly if exposed to an ambient environment and subsequent process steps. Without printable CGL materials, tandem architectures are not feasible.

Like any new technology, there is considerable risk involved in scaling the process from R&D/pilot production to high-volume mass production. Even though IJP OLED has been in development for many years, only small volumes of high-end monitors have been commercialized to date. The material systems are rapidly evolving, and the supply chain remains unproven. This contrasts with the long history of FMM OLED production, even with its known challenges. In the case of IJP, a variety of issues may only become obvious after a new Gen 8.6 factory is ramped up.

Cost comparison of IJP to FMM OLED

Detailed surveys of OLED supply chain companies, panel makers, and brands are useful research techniques to gather data, create lists of relevant information, and generate qualitative assumptions.

Such interviews return a widely accepted belief that IJP OLEDs should be lower cost than FMM OLEDs. These types of opinions, however, are typically not applied to specific examples and do not provide clear comparisons.

To confirm the validity of assumptions, put them in an appropriate context, and quantify the detailed cost differences, dedicated research and cost modeling were performed.

Results of this modeling are shown in Tables 4 and 5 below.

Table 4. 16.3" 3456x2234 hybrid IJP OLED for NB, 66-up on full G8.6 glass, circular polarizer, no color filter

Category	Subtotals	Details	2025	2026	2027	2028
	Yielded Array Material Cost		\$2.5	\$2.4	\$2.2	\$2.1
	RGB OLED		\$3.8	\$3.7	\$3.7	\$3.7
	IJP head replacement		\$0.2	\$0.2	\$0.2	\$0.2
	Yielded OLED Color Patterning Materials		\$7.6	\$7.2	\$6.7	\$6.3
	Yielded Encapsulation Material Cost		\$0.4	\$0.4	\$0.3	\$0.3
	Yielded Module Component Total		\$25.7	\$24.1	\$22.7	\$21.3
Labor Cost			\$4.7	\$5.0	\$5.2	\$5.5
Depreciation Cost (5 year example)			\$19.2	\$18.9	\$18.6	\$18.3
Indirect Expense Total			\$3.4	\$3.2	\$3.1	\$2.9
Manufacturing Total Cost			\$63.6	\$61.1	\$58.8	\$56.7
Total Yield			75%	78%	81%	84%

"Hybrid" refers to panels fabricated on glass substrates, with thin film encapsulation (TFE).

Even though, in some cases, it is advantageous to use a color filter to widen the color gamut, increase the viewing angle, and brightness, particularly for IJP OLEDs, "Circular polarizer, no color filter" means that neither of the panels analyzed includes a CF.

Both examples assume that a circular polarizer is applied to reduce ambient light reflections. That is, these assumptions are held constant in both examples and do not affect the manufacturing total cost comparison.

Table 5. 16.3" 3456x2234 hybrid FMM OLED for NB, 60-up on half G8.6 glass, circular polarizer, no color filter

Category	Subtotals	Details	2025	2026	2027	2028
	Yielded Array Material Cost		\$2.8	\$2.6	\$2.4	\$2.3
	RGB OLED		\$6.3	\$5.9	\$5.6	\$5.2
	IJP head replacement		\$0.0	\$0.0	\$0.0	\$0.0
	Yielded OLED Color Patterning Materials		\$11.0	\$9.9	\$9.0	\$8.1
	Yielded Encapsulation Material Cost		\$0.4	\$0.4	\$0.4	\$0.3
	Yielded Module Component Total		\$25.8	\$24.2	\$22.8	\$21.4
Labor Cost			\$5.5	\$5.7	\$6.0	\$6.3
Depreciation Cost (5 year example)			\$43.6	\$43.6	\$43.6	\$43.6
Indirect Expense Total			\$3.8	\$3.6	\$3.4	\$3.2
Manufacturing Total Cost			\$90.7	\$88.8	\$87.1	\$85.5
Total Yield			75%	78%	81%	84%

FMM mask, cleaning, and other related costs are integrated into the RGB OLED material cost line of the example in Table 5.

IJP print heads are consumable parts. Full replacement of heads is expensive, but depreciated over the large number of panels manufactured every year, the cost per panel is low.

IJP material utilization is much higher than FMM VTE, but IJP materials are still comparatively expensive. Material prices tend to decline as volume increases. Even so, it will take a substantial amount of time for IJP material demand to reach the volume required to push continuous price declines.

Chinese panel makers are rapidly expanding the domestic supply base of OLED organic materials and IJP inks to increase their material selection flexibility, to improve material supplier support and responsiveness, and to accelerate price reductions.

In the examples analyzed here, OLED materials account for a relatively smaller portion of the total manufacturing costs than depreciation.

IJP machines and the required open mask evaporator have high productivity and are relatively lower priced than FMM VTE equipment.

This means that the large majority of IJP's cost savings versus FMM OLED are from lower depreciation costs.

This analysis applies a five-year straight-line depreciation assumption. Different depreciation approaches may be used by various companies. Chinese FPD makers often use a seven-year depreciation schedule.

After depreciation is complete, the cost advantage of IJP will decline considerably, but will remain positive if all other parameters remain equal.

Finally, improved panelization—six additional 16.3-inch panels in this study—reduces IJP manufacturing total costs by about 10%.

4. Conclusion

Forecast to grow from 11.6 million units in 2025 to 54.1 million units in 2030, OLED notebook PC panels are currently one of the most promising segments of the display industry.

The excellent image quality and the thin, narrow bezel profile of OLED panels are highly attractive to PC brands. However, to displace LCDs and to meet, and then exceed, forecast market penetration, competitive cost performance is important.

This study found that 16.3-inch IJP OLED notebook panel manufacturing total costs will fall to \$56.7 by 2028. That is 34% lower than the \$85.5 modeled for a similar panel fabricated using VTE.

With the world's first Gen 8.6, high-volume IJP OLED factory under construction from the second half of 2025, the significant potential cost savings from IJP OLED manufacturing may finally be confirmed, as well as enable the further expansion of the OLED display market.

5. References

- [1] Cao W, IJP OLED Status and Future Prospects, TCL China Star. Omdia Korea Display Conference, October 2024.
- [2] Kwan HC, Noh, Y, Hwang JY. Control of Vacuum Pressure in Vacuum Drying Process for Uniform Film Formation of Inkjet-Printed OLEDs. Korea Institute of Industrial Technology. SID 2024 Digest; 1772-1774.
- [3] Yoshida H, Nakatani S, Inoue T, Usui Y, Ohtsuka F. 1pL Inkjet Head and G8.5 Equipment Development for 350ppi OLED Display Panels. Panasonic Production Engineering Corporation. SID 2025 Digest; 114-117.
- [4] Kim S, Kang T, Kim H, Mun H, Koo Y, Ha J, Lee C. All-Inkjet-printed AMOLED Display with Improved Efficiency and Lifetime. Display Research Center, Samsung Display Co., Ltd. SID 2022 Digest; 391-394.
- [5] Park R. Display Long-Term Demand Forecast Tracker. Omdia. October 2025.
- [6] Kim J, Uno T, Heo I. OLED Display Cost Model. Omdia. February 2025.